INCH-POUND

MIL-PRF-19500/667
26 June 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTOR, N-CHANNEL
SILICON TYPES 2N7456U1, 2N7457U1, 2N7458U1, AND 2N7459U1
JANTXVD, JANTXVR, JANSD, AND JANSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a N-Channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event characterization - see figure 4), power transistor. Two levels of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, SMD-1 (surface mount).

1.3 Maximum ratings. T_A = +25°C, unless otherwise specified.

Type	P_T (1) T_C = +25°C	P_T (1) T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} (2) T_C =+25°C	I_{D2} T_C = +100°C	I_S (2)	I_{DM}	T_J and T_{STG}	V_{ISO} 70,000 ft. altitude
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7456U1	150	60	100	100	±20	39	25	39	117	-55	N/A
2N7457U1	150	60	200	200	±20	27	17	27	81	to	N/A
2N7458U1	150	60	250	250	±20	21	14	21	63	+150	250
2N7459U1	150	60	500	500	±20	11	7	11	33		500

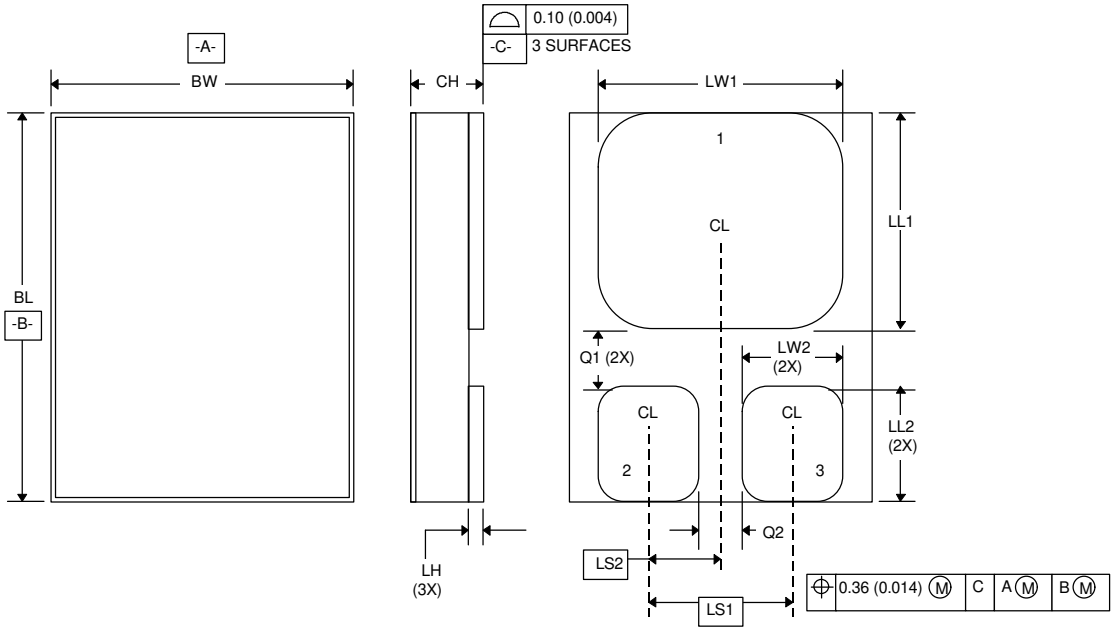
(1) Derate linearly 1.20 W/°C for T_C > +25°C; $P_T = (T_{jmax} - T_C)/R_{\theta JC}$
(2) $I_D = ((T_{jmax} - T_C)/((R_{\theta JC}) \text{ X } (r_{DS(on)} \text{ at } T_{jmax})))^{1/2}$.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC/VAC, Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	0.620	0.630	15.74	16.00
BW	0.445	0.455	11.30	11.55
CH	0.129	0.139	3.27	3.53
LH	0.010	0.020	0.25	0.51
LW1	0.370	0.380	9.39	9.65
LW2	0.135	0.145	3.43	3.68
LL1	0.410	0.420	10.41	10.66
LL2	0.152	0.162	3.86	4.11
LS1	0.200	0.220	5.08	5.59
LS2	0.100	0.110	2.54	2.79
Q1	0.035	---	0.89	---
Q2	0.050	---	1.27	---
TERM 1	Drain			
TERM 2	Gate			
TERM 3	Source			

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. The lid shall be electrically isolated from the Drain, Gate and Source.
4. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Dimensions and configuration .

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)1}$ V_{DS} = V_{GS} I_D = 1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{GS} = 80% of rated V_{DS}	Max $r_{DS(on)}$ (1) V_{GS} = 12V		R $_{\theta JC}$ Max	I_{AS} = I_{DM}
					T_J = +25°C at I_{D2}	T_J = +125°C at I_{D2}		
	V dc	V dc		µA dc	Ω	Ω	°C/W	A (pk)
		Min	Max					
2N7456U1	100	1.5	4.0	25	0.055	0.086	0.83	117
2N7457U1	200	1.5	4.0	25	0.100	0.182	0.83	81
2N7458U1	250	1.5	4.0	25	0.150	0.276	0.83	63
2N7459U1	500	1.5	4.0	25	0.530	1.02	0.83	33

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the test of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (SMD-1) herein. Methods used for electrical isolation of the terminals shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic).

3.4.1 Terminal material and finish. Terminal material shall be copper-tungsten. Terminal finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor but shall be retained on the initial container.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

3.8 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.8.1 Handling. MOSFET devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOSFET areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R ≤ or 100 kΩ, whenever bias voltage is applied drain to source.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein. Alternate flow is allowed for qualification inspection in accordance with figure 4 of MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table III herein. End point electrical measurements shall be in accordance with table I, group A, subgroup 2 herein (unless otherwise specified).

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4.3 Screening. Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS	JANTXV
(1)	Method 3470 E_{AS} test (see 4.5.4)	Method 3470 E_{AS} test (see 4.5.4)
(1)	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
(1)	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
(2)	Subgroup 2 of table I herein	Subgroup 2 of table I herein
9	I_{DSS1} , I_{GSS} as a minimum	Not applicable
10	MIL-STD-750, method 1042 test condition B	MIL-STD-750, method 1042 test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein.
12	MIL-STD-750, method 1042, test condition A, 240 hours minimum.	MIL-STD-750, method 1042 test condition A, 160 hours minimum.
13	Subgroups 2 and 3 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20% of initial value. $\Delta V_{GS(TH)1}$ = ±20% of initial value.	Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20% of initial value. $\Delta V_{GS(TH)1}$ = ±20% of initial value.

(1) Shall be performed anytime before screen 10.
(2) Shall be performed after E_{AS} test, method 3161, and gate stress test.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JANTXV) of MIL-PRF-19500 and paragraphs 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table IV herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles
B3	2077	SEM
B4	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink nor or forced-air cooling on the device shall be permitted during the on cycle. T_{on} = 30 seconds minimum.
B5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated; T_A = +175°C; t = 120 hours, min.
B5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated; T_A = +175°C; t = 24 hours.
B5	2037	Bond strength (Al-Au die interconnects only), test condition A
B6	3161	Thermal resistance, see 4.5.2.

4.4.2.2 Group B inspection, table VIb (JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. T_{on} = 30 seconds minimum.
B5	3161	Thermal resistance, see 4.5.2.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table IV herein.

Subgroup	Method	Condition
C2	2036	Not applicable.
C6	1042	Intermittent operation life, condition D, 6,000 cycles. No heat sink nor or forced-air cooling on the device shall be permitted during the on cycle. T_{on} = 30 seconds minimum.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table VIII of MIL-PRF-19500 and table II herein.

4.4.4.1 Design parameters. Not tested on a per lot basis. Design shall be such that the devices shall be capable of meeting the requirements on figure 4.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC}$ = 0.83°C/W. The following parameters shall apply:

a. Measuring current(I_M)..... 10 mA

b. Drain heating current (I_H)..... 4 A

c. Heating time (t_H)..... Steady-state (see MIL-STD-750, method 3161)

d. Drain-source heating voltage (V_H)..... 20V

e. Measurement time delay (t_{MD})..... 30 to 60 µs

f. Sample window time (t_{SW})..... 10 µs maximum

4.5.3 Thermal response ΔV_{SD} measurement). The delta V_{SD} measurement shall be performed in accordance with method 3161 of MIL-STD-750. The delta V_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 2). The chosen delta V_{SD} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report. The chosen delta V_{SD} shall be considered final after the manufacturer has had the opportunity to test three consecutive lots.

a. Measuring current (I_M)..... 10 mA

b. Drain heating current (I_H)..... 4 A

c. Heating time (t_H)..... 10 ms

d. Drain-source heating voltage (V_H)..... 25 V

e. Measurement time delay (t_{MD})..... 30 - 60 µs

f. Sample window time (t_{SW})..... 10 µs maximum

4.5.4 Single pulse avalanche energy (E_{AS}).

a. $I_{AS} = I_{DM}$

b. L = 0.1 mH

c. Gate to source resistor ($25\Omega \leq R_{GS} \leq 200\Omega$)

d. $E_{AS} = 1/2\ LI_{AS}^2$

e. V_{DD} = 50 V to 150 V dc

f. Initial junction temperature = +25°C, -5°C, +10°C

4.5.5 Gate stress test.

a. V_{GS} = 30 V

b. t = 250 µs, minimum

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage drain to source	3407	V_{GS} = 0V, I_D = 1 mA dc, Bias condition C	$V_{(BR)DSS}$			
2N7456U1				100		V dc
2N7457U1				200		V dc
2N7458U1				250		V dc
2N7459U1				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} = V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)1}$	1.5	4.0	V dc
Gate current	3411	V_{GS} = ±20V dc, Bias condition C, V_{DS} = 0V	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0V dc, Bias condition C, V_{DS} = 80 % of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source "ON" state resistance	3421	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)1}$			
2N7456U1					0.055	Ω
2N7457U1					0.100	Ω
2N7458U1					0.150	Ω
2N7459U1					0.530	Ω
Static drain to source "ON" state voltage	3405	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$	$V_{DS(ON)}$			
2N7456U1					2.34	V dc
2N7457U1					2.84	V dc
2N7458U1					3.31	V dc
2N7459U1					6.12	V dc
Forward voltage	4011	V_{GS} = 0V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D1}$	V_{SD}		1.8	V dc
Subgroup 3						
High temperature operation		$T_C = T_J$ = +125°C				
Gate current	3411	V_{GS} = ±20V dc, Bias condition C, V_{DS} = 0V	I_{GSS2}		±200	nA dc

See footnote at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 3 - continued.						
High temperature operation		$T_C = T_J = +125°C$				
Drain current	3413	V_{GS} = 0V dc, Bias condition C, V_{DS} = 80% of rated V_{DS}	I_{DSS2}		0.250	mA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)2}$			
2N7456U1					0.086	Ω
2N7457U1					0.182	Ω
2N7458U1					0.276	Ω
2N7459U1					1.02	Ω
Gate to source voltage (threshold)	3403	$V_{DS} = V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)2}$	0.5		V dc
Low temperature operation		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} = V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = 12 V dc, R_G = 2.35 Ω,				
Turn-on delay time		V_{DD} = 50% of rated V_{DS}	$t_{D(on)}$			
2N7456U1					40	ns
2N7457U1					45	ns
2N7458U1					40	ns
2N7459U1					40	ns
Rise Time			t_R			
2N7456U1					130	ns
2N7457U1					110	ns
2N7458U1					65	ns
2N7459U1					20	ns
Turn-off delay time			$t_{D(off)}$			
2N7456U1					75	ns
2N7457U1					90	ns
2N7458U1					80	ns
2N7459U1					85	ns
Fall time			t_f			
2N7456U1					35	ns
2N7457U1					25	ns
2N7458U1					15	ns
2N7459U1					15	ns

See footnote at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area test (high voltage)	3474	See figure 3, t_p = 10 ms min. V_{DS} = 80% of max. rated V_{DS} ($V_{DS} \leq$ 200 V)				
Electrical measurements		Table I, group A, subgroup 2 herein.				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition A or B				
On-state gate charge			$Q_{G(ON)}$			
2N7456U1					130	ns
2N7457U1					150	ns
2N7458U1					150	ns
2N7459U1					130	ns
Gate to source charge			Q_{GS}			
2N7456U1					28	ns
2N7457U1					40	ns
2N7458U1					36	ns
2N7459U1					26	ns
Gate to drain charge			Q_{GD}			
2N7456U1					75	ns
2N7457U1					78	ns
2N7458U1					76	ns
2N7459U1					62	ns
Reverse recovery time	3473	di/dt = 100 A/µs, $V_{DD} \leq$ 50 V, $I_D = I_{D1}$	t_{rr}			
2N7456U1					530	ns
2N7457U1					550	ns
2N7458U1					820	ns
2N7459U1					980	ns

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/ 4/	MIL-STD-750		Symbol	Pre-irradiation limits		Post irradiation limits		Units
	Method	Conditions		Min.	Max.	Min.	Max.	
Subgroup 1								
Not Applicable								
Subgroup 2		T_C = +25°C						
Steady state total dose irradiation (V_{GS} bias)	1019	V_{GS} = 12V, V_{DS} = 0V						
Steady state total dose irradiation (V_{DS} bias)	1019	V_{GS} = 0V, V_{DS} = 80% of rated V_{DS}						
Breakdown voltage drain to source	3407	V_{GS} = 0V, I_D = 1 mA dc, Bias condition C	$V_{(BR)DSS}$					
2N7456U1				100		100		V dc
2N7457U1				200		200		V dc
2N7458U1				250		250		V dc
2N7459U1				500		500		V dc
Gate to source Voltage (threshold)	3403	V_{DS} = V_{GS}, I_D = 1 mA dc	$V_{GS(TH)1}$	1.5	4.0	1.5	4.0	V dc
Gate current	3411	V_{GS} = ±20V dc, V_{DS} = 0V, Bias condition C	I_{GSS1}		±100		±100	nA dc
Drain current	3413	V_{GS} = 0V, V_{DS} = 80% of rated V_{DS}, bias condition C	I_{DSS1}		25		25	µA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 12V dc, condition A, pulsed (see4.5.1), I_D = I_{D2}	$r_{DS(ON)1}$					
2N7456U1					0.055		0.055	Ω
2N7457U1					0.100		0.100	Ω
2N7458U1					0.150		0.150	Ω
2N7459U1					0.530		0.530	Ω
Static drain to source "ON"-state voltage	3405	V_{GS} = 12V dc, condition A, Pulsed (see 4.5.1), I_D = I_{D1}	$V_{DS(ON)}$					
2N7456U1					2.34		2.34	V dc
2N7457U1					2.84		2.84	V dc
2N7458U1					3.31		3.31	V dc
2N7459U1					6.12		6.12	V dc

1/ For sampling plan see MIL-PRF-19500.
2/ Electrical specifications are for 'D' and 'R' rad levels.
3/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other detail specification utilizing the same die design.
4/ At the manufacturer's option, group D samples need not be subjected to all the screening tests, but shall be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.

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TABLE III. Group E inspection (all quality levels) - for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection.
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition G, 500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		Table I, group A, subgroup 2 herein.	
Subgroup 2 1/			12 devices c = 0
Steady-state gate bias	1042	Test condition B; 1,000 hours	
Electrical measurements		Table I, group A, subgroup 2 herein.	
Steady state reverse bias	1042	Test condition A; 1,000 hours	
Electrical measurements		Table I, group A, subgroup 2 herein.	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 0.83°C/W maximum. See 4.5.2	
Subgroup 5			15 devices c = 0
Barometric pressure test (not required for $V_{BR(DSS)} \leq 200V$)	1001	Test condition C	
2N7458U1 2N7459U1		V_{DS} = 250 V; $I_{(ISO)}$ < 0.25 mA V_{DS} = 500 V; $I_{(ISO)}$ < 0.25 mA	

See footnotes at end of table.

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TABLE III. Group E inspection (all quality levels) - for qualification only - Continued.

Inspection	MIL-STD-750		Qualification and large lot conformance inspection
	Method	Conditions	
Subgroup 6 2/			3 devices 5/ c = 0
Electrical measurements 3/		I_{GSS1} and I_{DSS1} of table I, group A, subgroup 2herein.	
SEE Effect Testing 4/	1080	Fluence = 3e5 ± 20 percent ions/cm^2 Flux = 5e3 to 2e4 ions/cm^{2}sec Beam energy = 260 to 300 MeV Temperature = +25°C ± 5°C	
2N7456U1		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 100 V and V_{GS} = -20 V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 100 V and V_{GS} = -10 V V_{DS} = 80 V and V_{GS} = -15 V V_{DS} = 50 V and V_{GS} = -20 V	
2N7457U1		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 200 V and V_{GS} = -20 V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 200 V and V_{GS} = -5 V V_{DS} = 160 V and V_{GS} = -10 V V_{DS} = 100 V and V_{GS} = -15 V V_{DS} = 40 V and V_{GS} = -20 V	
2N7458U1		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 250 V and V_{GS} = -20 V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 250 V and V_{GS} = -5 V V_{DS} = 200 V and V_{GS} = -10 V V_{DS} = 125 V and V_{GS} = -15 V V_{DS} = 50 V and V_{GS} = -20 V	
2N7459U1		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = 500 V and V_{GS} = -15 V V_{DS} = 450 V and V_{GS} = -20 V LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = 500 V and V_{GS} = -5 V V_{DS} = 400 V and V_{GS} = -10 V V_{DS} = 100 V and V_{GS} = -15 V	

See footnotes at end of table.

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TABLE III. Group E inspection (all quality levels) - for qualification only - Continued.

Inspection	MIL-STD-750		Qualification and large lot conformance inspection
	Method	Conditions	
Subgroup 6 2/			3 devices 5/ c = 0
Electrical measurements 3/		I_{GSS1} and I_{DSS1} of table I, group A, subgroup 2 herein.	
ESD (Electrostatic Discharge Protection)	1020		

1/ A separate sample for each test may be pulled.
2/ Group E qualification of single event effect testing may be performed prior to lot formation. Wafers qualified to these group E QCI requirements may be used for any other performance specification utilizing the same die design.
3/ As a minimum, gate to source leakages and drain to source leakage are to be examined to verify the electrical performance of the DUT prior to and after test. At the manufacturer's option, the remaining static tests in table I, group A, subgroup 2, may be performed.
4/ Devices passing a given combination of drain and gate voltage for an LET of 36 to 40 MeV-cm^2/mg qualify the same conditions for an LET of 26 to 30 MeV-cm^2/mg.
5/ This sampling plan applies to each bias condition defined.

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TABLE IV. Group B, C, and E delta measurements.

Step	Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1.	Thermal response	3161	See 4.5.3	ΔV_{SD}		70	mV

1/ The delta V_{SD} measurements for table VIa (JANS) of MIL-PRF-19500 are as follows: Subgroup 4, see table IV herein, step 1.

2/ The delta V_{SD} measurements for table VIb (JANTXV) of MIL-PRF-19500 are as follows: Subgroup 3, see table IV herein, step 1.

3/ The delta V_{SD} measurements for table VII of MIL-PRF-19500 are as follows: Subgroup 6, see table IV herein, step 1.

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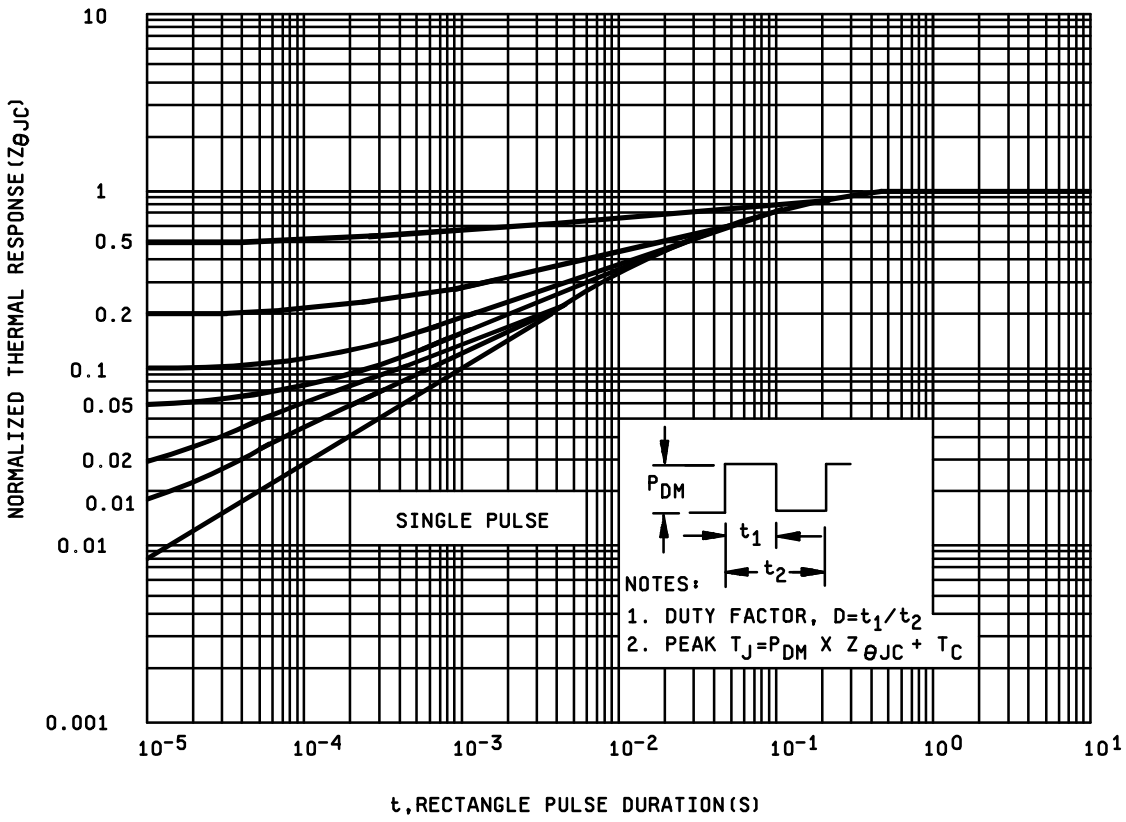


FIGURE 2. Thermal response curves.

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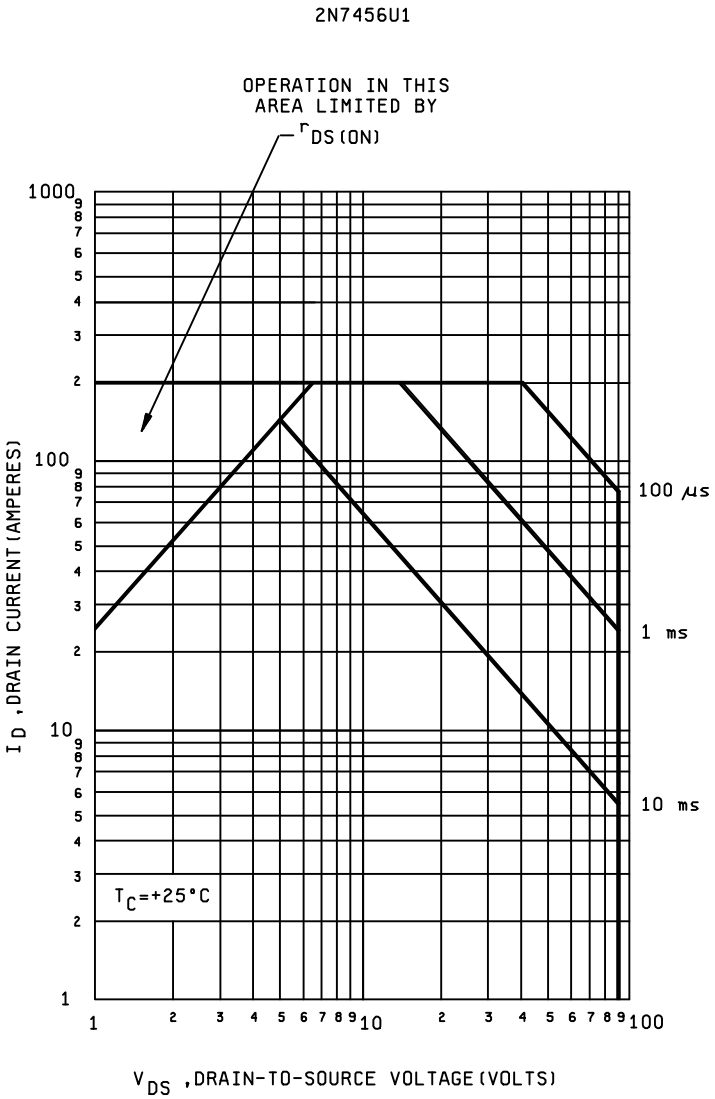


FIGURE 3. Safe operating area graphs.

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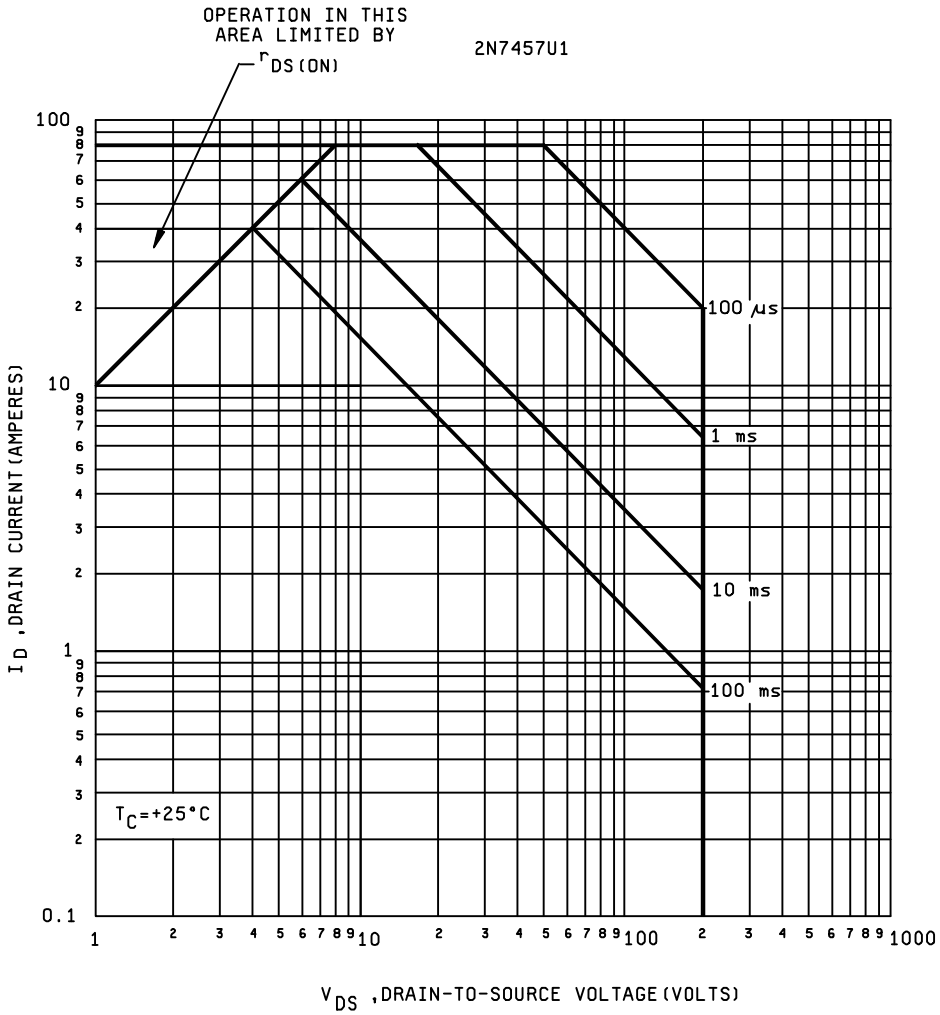


FIGURE 3. Safe operating area graphs – Continued.

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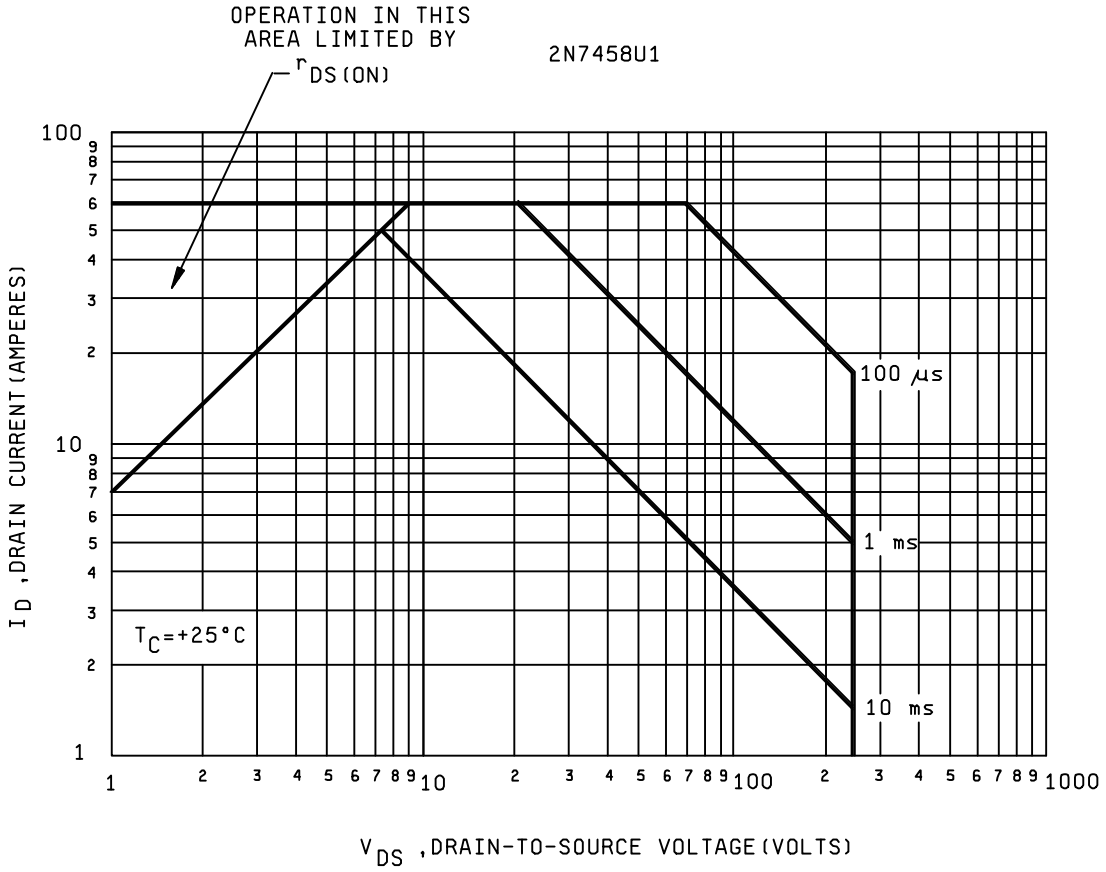


FIGURE 3. Safe operating area graphs – Continued.

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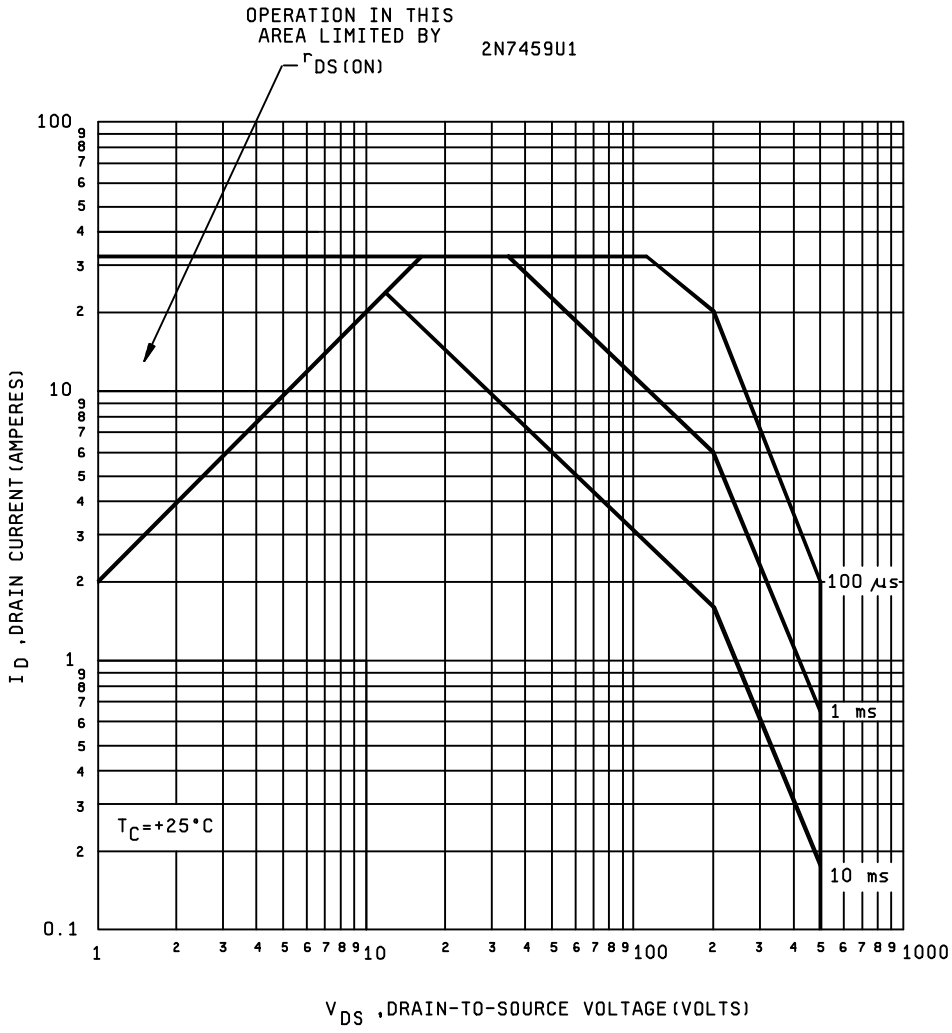


FIGURE 3. Safe operating area graphs – Continued.

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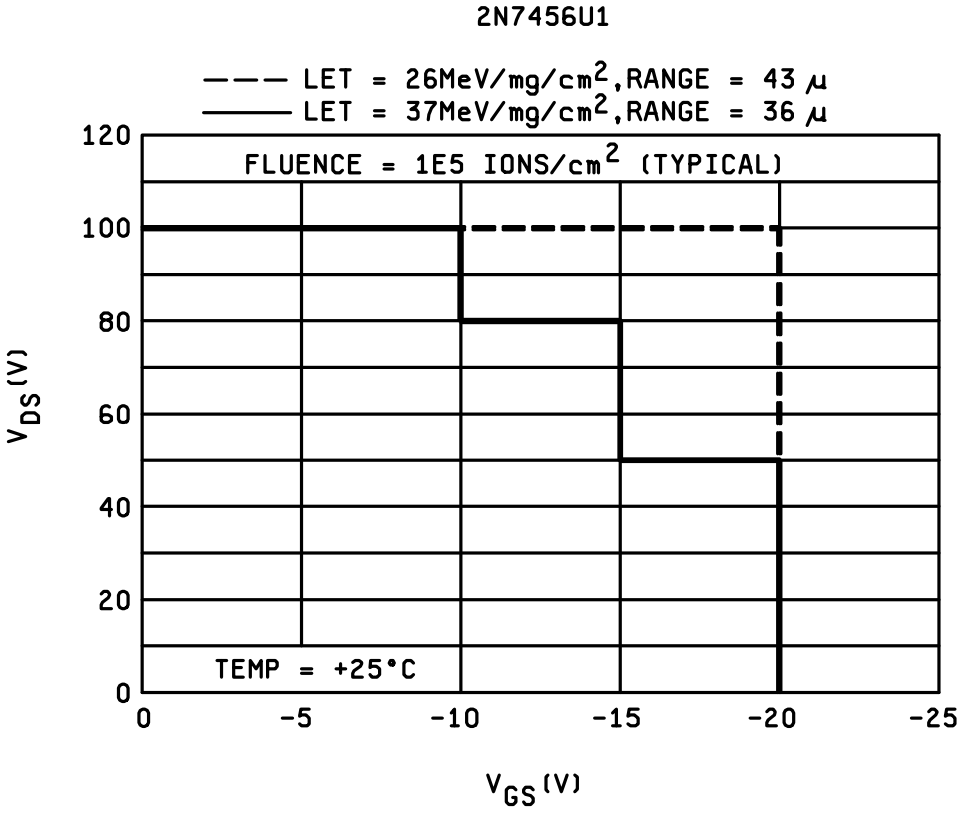


FIGURE 4. Single event effects safe operating area graphs.

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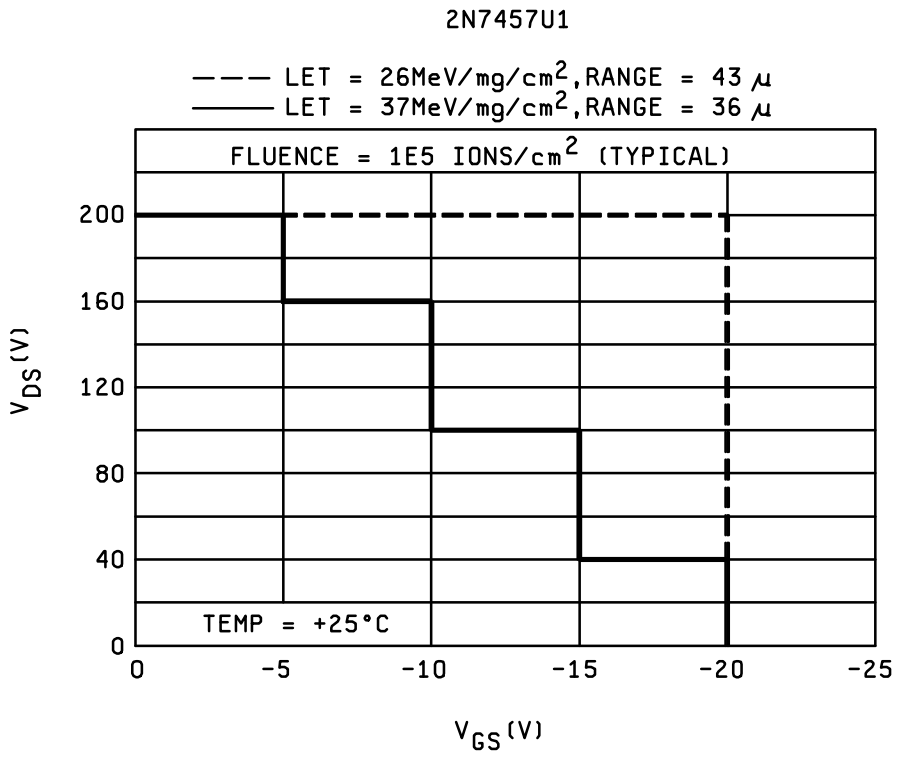


FIGURE 4. Single event effects safe operating area graphs – Continued.

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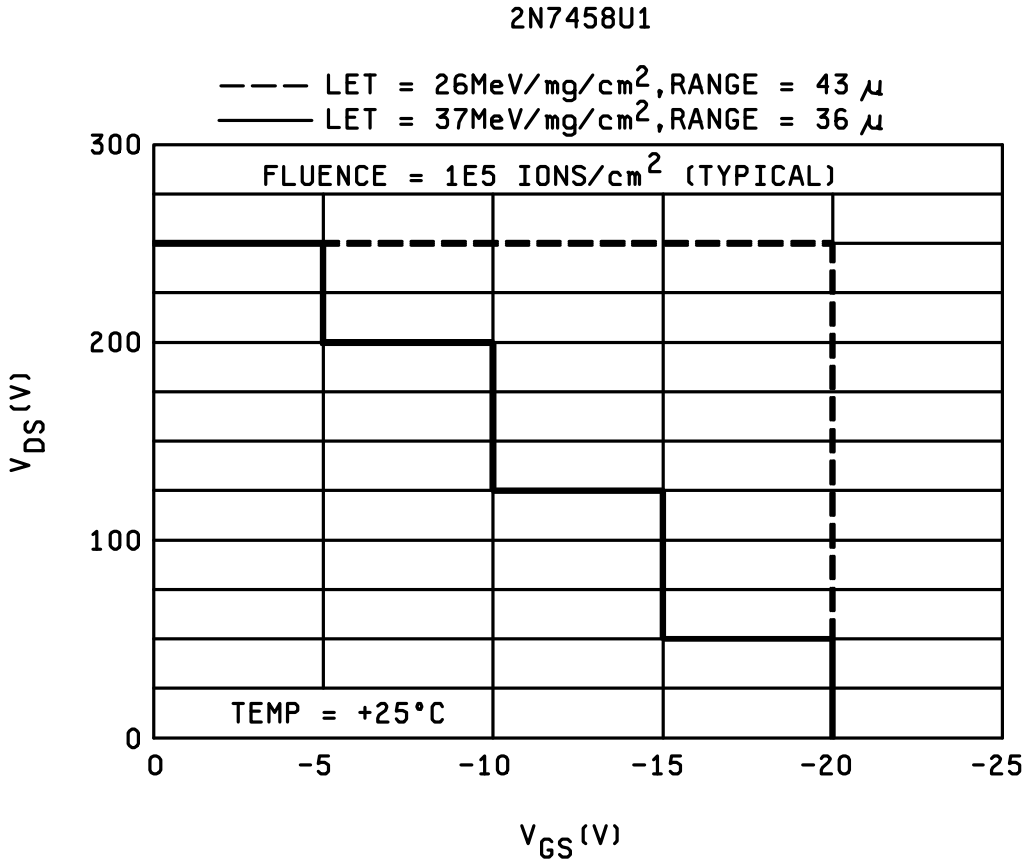


FIGURE 4. Single event effects safe operating area graphs – Continued.

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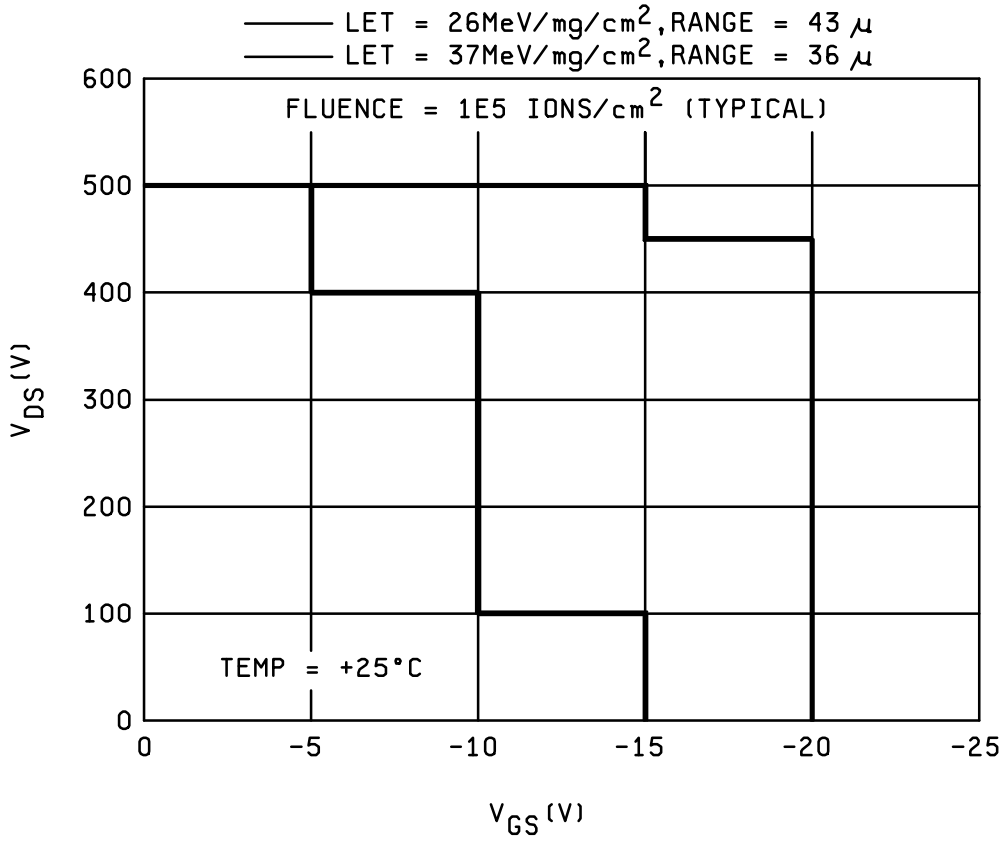


FIGURE 4. Single event effects safe operating area graphs – Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. The acquisition requirements shall be as specified in MIL-PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Cross-reference list. The following table shows the generic PIN (part identifying number) and its associated military PIN (without JAN and RHA prefix).

Generic P/N	Military P/N
SYA150	2N7456U1
SYA250	2N7457U1
SYA254	2N7458U1
SYA450	2N7459U1

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2122)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/667	2. DOCUMENT DATE 26 June 2000

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTOR, N-CHANNEL SILICON TYPES 2N7456U1U1, 2N7457U1, 2N7458U1U1, AND 2N7459U1U1 JANTXVD, JANTXVR, JANSD, AND JANSR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0510 850-0510 614-692-6939 alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43213-1199	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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